HSBI — Bielefeld University of Applied Sciences and Arts

Quantitative Research Methods

Summer Semester 2026 · Examiner: Prof. Dr. Christoph Weisser

Mock Examination 2 (Lectures 5–8)

ISLP Chapters 4–6: Classification · Resampling Methods · Model Selection and Regularisation

Duration: **90 minutes**
Total credit: **90 points**
Allowed aids: non-programmable calculator; one handwritten A4 sheet of notes (both sides)

Name: ______________________________ Matriculation no.: ____________________

Problem	**P1**	**P2**	**P3**	**P4**	**P5**	**P6**	**Total**
Maximum points	18	12	15	18	12	15	**90**
Achieved points							

Instructions

- Justify all answers. Numerical results without a derivation or a brief reasoning receive no credit.
- Round final numerical results to 3 decimal places unless stated otherwise. Small deviations caused by carrying rounded intermediate results are accepted.
- The number of points per problem roughly equals the recommended working time in minutes.

Problem 1: Logistic regression by hand (18 points)

A bank models the probability that a credit-card customer *defaults* ($Y = 1$) as a function of the monthly card balance x (in EUR; balances in this portfolio range from 0 to 400 EUR). A logistic regression fitted by maximum likelihood gave

$$\hat{p}(x) = \Pr(Y = 1 \mid X = x) = \frac{e^{\hat{\beta}_0+\hat{\beta}_1 x}}{1 + e^{\hat{\beta}_0+\hat{\beta}_1 x}}, \qquad \hat{\beta}_0 = -4, \quad \hat{\beta}_1 = 0.02.$$

(a) **[4 P]** Compute the predicted default probability $\hat{p}(x)$ for a customer with balance $x = 100$ EUR and for a customer with balance $x = 250$ EUR.

(b) **[4 P]** For the same two customers, compute the *odds* of default $\hat{p}(x)/(1-\hat{p}(x))$ and the *log-odds* (logit). Interpret the odds of the customer with $x = 250$ in one sentence.

(c) **[3 P]** Determine the balance x^* at which the predicted default probability equals exactly 0.5. Show your derivation.

(d) **[4 P]** Interpret the estimated slope on the odds scale: by which *factor* do the estimated odds of default change (i) per additional EUR of balance and (ii) per additional 100 EUR of balance? Why is the statement "$\hat{\beta}_1 = 0.02$ means the default *probability* rises by 0.02 per EUR" wrong?

(e) **[3 P]** Give two reasons why an ordinary linear regression of the 0/1 response Y on x would be inappropriate here, even though it can be computed mechanically.

Problem 2: Discriminant analysis and naive Bayes — concepts (12 points)

(a) **[4 P]** Linear discriminant analysis (LDA) and quadratic discriminant analysis (QDA) both model the class-conditional densities $f_k(x)$ of the predictors as multivariate normal. State precisely the assumption about the covariance matrices that distinguishes the two methods, and give the number of covariance *matrices* each method estimates when there are K classes.

(b) **[3 P]** Using the bias–variance trade-off, explain in which situations LDA tends to outperform QDA on test data, and in which situations QDA tends to win. Mention the role of the sample size n and of the true class covariances.

(c) **[3 P]** State the key assumption of the *naive Bayes* classifier about the predictors within each class, and explain why this assumption can improve test performance when the number of predictors p is large relative to n — even though the assumption is rarely exactly true.

(d) **[2 P]** Which of the three classifiers (LDA, QDA, naive Bayes) produce *linear* decision boundaries in x, and which produce *quadratic* or more general non-linear boundaries?

Problem 3: Confusion matrix and ROC curve (15 points)

A logistic regression classifier predicts credit-card default on an evaluation set of $n = 1{,}000$ customers, using the default threshold: predict "default" if $\hat{p}(\text{default} \mid x) > 0.5$. The result is the following confusion matrix:

		True status		
		Default	No default	Total
Predicted	Default	60	45	105
	No default	40	855	895
	Total	100	900	1,000

(a) **[7 P]** Identify TP, FP, FN and TN, and compute (i) the overall accuracy and the overall error rate, (ii) the sensitivity (recall, true positive rate), (iii) the specificity, (iv) the precision, and (v) the false positive rate. Interpret sensitivity and precision in one sentence each in the context of this application.

(b) **[4 P]** The bank now lowers the classification threshold from 0.5 to 0.2 (predict "default" if $\hat{p} > 0.2$). Explain — without further computation — how sensitivity and specificity will change, and give a business reason why the bank might deliberately accept this trade-off.

(c) **[4 P]** The ROC curve plots which two quantities against each other, as what is varied? What does an area under the curve (AUC) of 0.5 mean, and what does an AUC of 1 mean?

Problem 4: Resampling methods (18 points)

(a) **[4 P]** A data set has $n = 200$ observations. To estimate the test error of a fixed model specification, state how many times the model must be *fitted* for (i) the validation-set approach (one 50/50 split), (ii) 10-fold cross-validation, and (iii) leave-one-out cross-validation (LOOCV), with a one-sentence justification each.

(b) **[4 P]** Compare LOOCV and 10-fold CV as estimators of the true test error with respect to *bias* and *variance*. Which method is preferred in practice, and why?

(c) **[5 P]** A bootstrap sample of size n is drawn with replacement from n observations. Show that the probability that a fixed observation i is *contained* in the bootstrap sample equals $1-(1-1/n)^n$. Evaluate this probability for $n=5$, and state the limiting value as $n \to \infty$ (use $e^{-1} \approx 0.368$).

(d) **[5 P]** To quantify the uncertainty of an estimated portfolio weight $\hat{\alpha}$, an analyst draws $B=3$ bootstrap samples (in practice $B \approx 1{,}000$; $B=3$ is used here only to keep the arithmetic manageable) and obtains

$$\hat{\alpha}^{*1} = 0.4, \qquad \hat{\alpha}^{*2} = 0.5, \qquad \hat{\alpha}^{*3} = 0.6.$$

Compute the bootstrap estimate of the standard error,

$$\widehat{\mathrm{SE}}_B(\hat{\alpha}) = \sqrt{\frac{1}{B-1}\sum_{r=1}^{B}\Big(\hat{\alpha}^{*r} - \bar{\alpha}^{*}\Big)^2}, \qquad \bar{\alpha}^{*} = \frac{1}{B}\sum_{r=1}^{B}\hat{\alpha}^{*r},$$

and explain in one sentence what this number estimates. Why is the bootstrap useful here although a formula for $\mathrm{SE}(\hat{\alpha})$ may be hard to derive?

Problem 5: Model selection with C_p and BIC (12 points)

Best subset selection on a data set with $n = 100$ observations produced, for each model size d (number of predictors), the best model and its residual sum of squares:

Number of predictors d	1	2	3	4
RSS_d	500	444	428	424

The noise variance is estimated from the full model as $\hat{\sigma}^2 = 4$. Use the criteria

$$C_p = \frac{1}{n}\big(\mathrm{RSS} + 2\, d\, \hat{\sigma}^2\big), \qquad \mathrm{BIC} = \frac{1}{n}\big(\mathrm{RSS} + \log(n)\, d\, \hat{\sigma}^2\big), \qquad \log(100) \approx 4.605.$$

(a) **[4 P]** Compute C_p for all four models (3 decimals).

(b) **[4 P]** Compute BIC for all four models (3 decimals).

(c) **[2 P]** Which model size does each criterion select? Explain the reason for the disagreement by comparing the two penalty terms.

(d) **[2 P]** Why can the training RSS (or equivalently the training R^2) alone never be used to select d? What does RSS necessarily do as d grows?

Problem 6: Ridge regression and the lasso (15 points)

Consider a linear model with response y and p standardised predictors $X_1, \ldots, X_p$. Part (e) also draws on the bias–variance ideas of Chapter 2 and the least-squares fit of Chapter 3.

(a) **[4 P]** Write down the optimisation objectives of *ridge regression* and of the *lasso* (RSS plus the respective penalty, with tuning parameter $\lambda \geq 0$). Which coefficient is conventionally *not* penalised?

(b) **[3 P]** Describe the behaviour of the two estimators in the limits $\lambda \to 0$ and $\lambda \to \infty$. Which method from Chapter 3 is recovered as $\lambda \to 0$?

(c) **[3 P]** Explain — in words, using the constraint-region geometry for $p = 2$ — why the lasso can set coefficients *exactly* to zero while ridge regression typically does not.

(d) **[2 P]** Why should the predictors be *standardised* before ridge or lasso is applied, when this is irrelevant for ordinary least squares?

(e) **[3 P]** A ridge regression was fitted for two values of the tuning parameter; the table shows the standardised coefficient estimates. A 10-fold cross-validation over a grid of λ values gave the CV errors on the right.

	$\lambda = 0.1$	$\lambda = 100$
$\hat{\beta}_1$	0.92	0.31
$\hat{\beta}_2$	0.85	0.29
$\hat{\beta}_3$	−0.41	−0.12
$\hat{\beta}_4$	0.20	0.06

λ	0.1	1	10	100
CV error	12.8	11.9	10.6	13.4

Which of the two fits shown in the coefficient table has the lower bias and which the lower variance, and why? Which λ from the CV table should be chosen for the final model?

End of the examination — good luck!